# Cooling capacity and SHR

24 unit-specific curve pairs fitted to 1,432 valid catalog points, using the largest listed airflow for each size. 18 manufacturer blanks remain unavailable. The fitted inputs are outdoor dry bulb, entering wet bulb and entering dry bulb; airflow is fixed, not an input.

## Three temperature inputs, one fixed airflow selection

Separate fits retain the catalog size and efficiency labels. Selected airflow is 480 CFM per nominal ton in every included table. Source SHR is sensible cooling capacity divided by total gross cooling capacity. Capacity is gross, before indoor fan heat, following the source notes.

## Equations - all temperatures in degrees Fahrenheit

```
x = (outdoor DB - 95) / 20

y = (entering WB - 67) / 6

z = (entering DB - 80) / 10

F = 1 + c1*x + c2*y + c3*z + c4*x*x + c5*x*y

+ c6*x*z + c7*y*y + c8*y*z + c9*z*z

Gross capacity (Btu/hr) = Q_reference * F

SHR = min(1, max(0, (b0 + b1*x + b2*y + b3*z) / F))
```

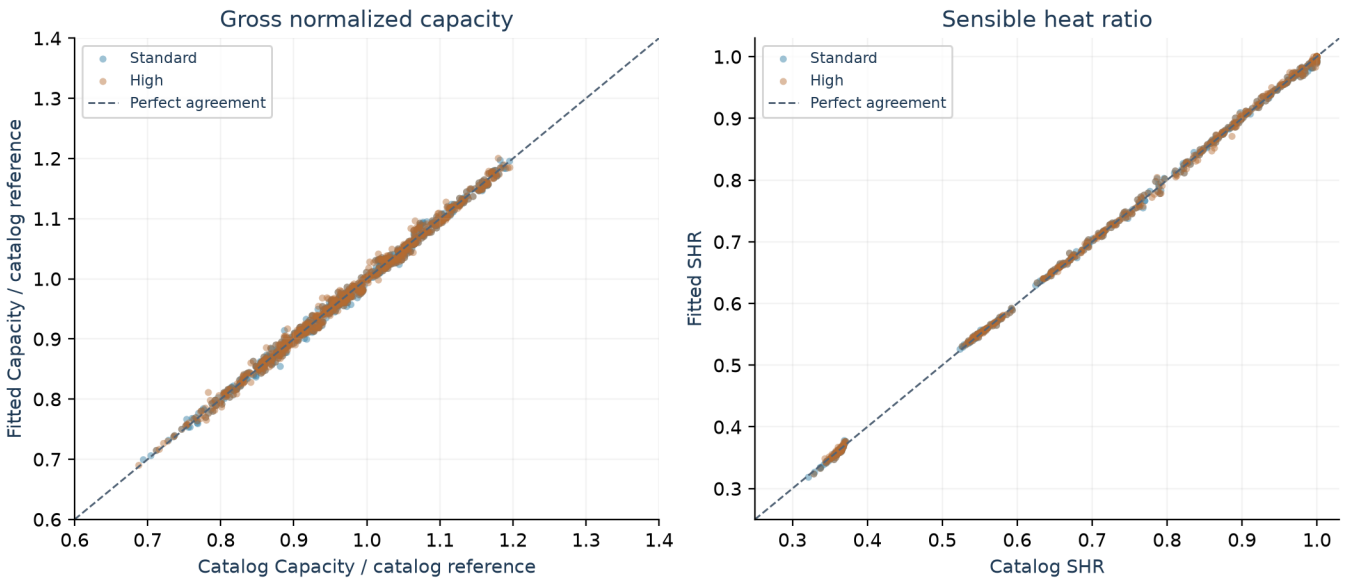
Q_reference is each unit's catalog capacity at 95 F outdoor DB, 67 F entering WB and 80 F entering DB, at the selected maximum airflow. It is neither nominal tonnage times 12,000 nor the peak building load. Capacity is normalized to exactly 1 at that reference. The four b coefficients are part of the SHR function; they introduce no additional operating input.

## Where the functions can be used

Entering WB: 61-73 F. Entering DB: 75-90 F. Outdoor DB: generally 85-125 F; standard 10-ton heat pump: 85-120 F. For both 25-ton heat-pump fits, also require outdoor DB + 2 x entering DB <= 295. These limits describe the convex hull of valid catalog points. No extrapolation is allowed.

Coefficients, catalog references, exact domain checks, source rows and validation are provided in the accompanying CSV/JSON files and standalone evaluator. Capacity and SHR fitting is separate from the cooling power and fan balance in the engineering guide.

# Fit accuracy and validation




Each marker compares one catalog point with its fitted result. Dashed lines show perfect agreement. Standard and high-efficiency points often overlap because several source datasets are identical.

| Measure | Range across fitted units |
|---|---|
| Capacity RMSE / reference capacity | 0.62% to 1.27% |
| SHR RMSE (absolute ratio) | 0.0014 to 0.0047 |
| Interior-plane validation: capacity RMSE / reference | 0.66% to 1.40% |
| Interior-plane validation: SHR RMSE | 0.0016 to 0.0061 |

## Validation method

Capacity uses anchored ordinary least squares. SHR uses bounded-output least squares for its numerator, with the fitted capacity as denominator. Interior-plane validation removes each interior outdoor-DB or entering-DB temperature plane, refits the model, and predicts the omitted points. The reference capacity stays known. Only three wet-bulb levels are supplied, so independent wet-bulb-plane validation is not claimed. These errors measure catalog agreement, not field performance.

Source: supplied cooling performance tables; heat-pump tables 10-33 only; dual-fuel tables 34-57 excluded. The selected maximum-airflow data retain PDF and printed page references, manufacturer blanks and accepted fitting values. Source hash is recorded in curve-fits.json.

# Unit-by-unit fit summary

## Standard efficiency - heat pump

| Tons | Fixed CFM | Qref MBh | Cap. RMSE % | SHR RMSE | Table |
|---|---|---|---|---|---|
| 3 | 1,440 | 38.9 | 0.75 | 0.0024 | 10 |
| 4 | 1,920 | 52.1 | 0.67 | 0.0025 | 11 |
| 5 | 2,400 | 61.9 | 0.65 | 0.0017 | 12 |
| 6 | 2,880 | 79.3 | 0.63 | 0.0041 | 13 |
| 7.5 | 3,600 | 92.0 | 0.67 | 0.0014 | 14 |
| 8.5 | 4,080 | 110.7 | 0.62 | 0.0025 | 15 |
| 10 | 4,800 | 124.1 | 1.07 | 0.0047 | 16 |
| 12.5 | 6,000 | 156.9 | 0.77 | 0.0029 | 17 |
| 15 | 7,200 | 189.6 | 0.74 | 0.0028 | 18 |
| 17.5 | 8,400 | 215.9 | 0.78 | 0.0042 | 19 |
| 20 | 9,600 | 253.5 | 0.79 | 0.0036 | 20 |
| 25 | 12,000 | 277.8 | 0.64 | 0.0024 | 21 |

## High efficiency - heat pump

| Tons | Fixed CFM | Qref MBh | Cap. RMSE % | SHR RMSE | Table |
|---|---|---|---|---|---|
| 3 | 1,440 | 37.8 | 1.27 | 0.0045 | 22 |
| 4 | 1,920 | 51.4 | 0.84 | 0.0033 | 23 |
| 5 | 2,400 | 62.9 | 0.82 | 0.0031 | 24 |
| 6 | 2,880 | 79.3 | 0.63 | 0.0041 | 25 |
| 7.5 | 3,600 | 92.0 | 0.67 | 0.0014 | 26 |
| 8.5 | 4,080 | 110.7 | 0.62 | 0.0025 | 27 |
| 10 | 4,800 | 124.1 | 0.78 | 0.0040 | 28 |
| 12.5 | 6,000 | 156.9 | 0.77 | 0.0029 | 29 |
| 15 | 7,200 | 189.6 | 0.74 | 0.0028 | 30 |
| 17.5 | 8,400 | 215.9 | 0.78 | 0.0042 | 31 |
| 20 | 9,600 | 253.5 | 0.79 | 0.0036 | 32 |
| 25 | 12,000 | 277.8 | 0.64 | 0.0024 | 33 |

Capacity RMSE is percent of the catalog reference, not percent of each individual data point. SHR RMSE is an absolute ratio. Per-point errors, relative-capacity errors and held-out predictions are supplied in the CSV files.